

Name: make_circle

Input:

- point0x: a real number representing the x-coordinate of the first point
- point0y: a real number representing the y-coordinate of the first point
- point1x: a real number representing the x-coordinate of the second point
- point1y: a real number representing the y-coordinate of the second point
- point2x: a real number representing the x-coordinate of the third point
- point2y: a real number representing the y-coordinate of the third point

Output: returns intersect_x, a real number representing the x coordinate of the center, intersect_y, a real number representing the y coordinate of the center, radius, a real number representing the radius of the circle

1. **midpoint1_x** $\leftarrow$ the first value returned from *midpoint*(point0x, point0y, point1x, point1y)
2. **midpoint1_y** $\leftarrow$ the second value returned from *midpoint*(point0x, point0y, point1x, point1y)
3. **midpoint2_x** $\leftarrow$ the first value returned from *midpoint*(point1x, point1y, point2x, point2y)
4. **midpoint2_y** $\leftarrow$ the second value returned from *midpoint*(point1x, point1y, point2x, point2y)
5. **perp_1** $\leftarrow$ the value returned from *perp*(slope(point0x, point0y, point1x, point1y))
6. **perp_2** $\leftarrow$ the value returned from *perp*(slope(point1x, point1y, point2x, point2y))
7. **intersect_x** $\leftarrow$ the first value returned from *intersect*(perp_1, midpoint1_x, midpoint1_y, perp_2, midpoint2_x, midpoint2_y)
8. **intersect_y** $\leftarrow$ the second value returned from *intersect*(perp_1, midpoint1_x, midpoint1_y, perp_2, midpoint2_x, midpoint2_y)
9. **radius** $\leftarrow$ the value returned from *distance*(intersect_x, intersect_y, point0x, point0y)
10. return intersect_x, intersect_y, radius

Discussion:

A more efficient strategy to find the center and radius of the circle with three arbitrary points on its circumference is to use the equation of a circle $(x - h)^2 + (y - k)^2 = r^2$, where the center is (h, k) and the radius is r . If we plug in the values of x and y for each of the three points, we have three equations, which are sufficient to find three unknowns. We can then use a system of equations to solve for h , k , and r .